INPUT AND VOLUME (RV1/RV2 = one dual pot, off-board, plugs into J2; 2 x X9C104 later on the same pins)

J1 LINE IN L/GND/R — INL, GND, INR

RV1 B100K lin gang A — INL, WL, GND

R3 10k — WL, GND

C13 47u 50V — W1, WL

R5 5.6k — W1, IN1

J2 VOLUME / CONTROL — INL, WL, INR, WR, GND, MUTE_CTL

RV2 B100K lin gang B — INR, WR, GND

R4 10k — WR, GND

C14 47u 50V — W2, WR

R6 5.6k — W2, IN2

LA4440 STEREO CORE (gain 30 dB: R7/R8 470R in the feedback legs)

U1 LA4440 — VCC; IN1, NF1, IN2, NF2, MUTE, DC; BS1, OUT1, BS2, OUT2; GND, GND2, GND1, PGND

R7 470 — NF1, NFC1

R8 470 — NF2, NFC2

C1 47u 50V — NFC1, GND

C2 47u 50V — NFC2, GND

C9 1000u 25V — DC, GND

C3 1000u 25V — BS1, OUT1

C4 1000u 25V — BS2, OUT2

OUTPUTS (Zobel = 0.1u film + 2 x 10R 1W in parallel)

C7 2200u 25V — OUT1, SPK1

C5 0.1u film — OUT1, ZB1

R14 10R 1W (R1 pair) — ZB1, GND

R15 10R 1W (R1 pair) — ZB1, GND

J5 SPEAKER L — SPK1, GND

LS1 4-8R — SPK1, GND

C8 2200u 25V — OUT2, SPK2

C6 0.1u film — OUT2, ZB2

R16 10R 1W (R2 pair) — ZB2, GND

R17 10R 1W (R2 pair) — ZB2, GND

J6 SPEAKER R — SPK2, GND

LS2 4-8R — SPK2, GND

MUTE: SW1 (off-board, J3) or MUTE_CTL high (GPIO) — wire-OR

J3 MUTE SWITCH — VCC, MV

SW1 MTS-112 MUTE — VCC, MV

R10 47k — VCC, PB

Q2 2N5401 — PB, VCC, MV

R9 1.8k — MV, MUTE

C10 47u 50V — MUTE, GND

R11 10k — PB, Q1C

Q1 2N3904 — Q1B, Q1C, GND

R12 4.7k — MUTE_CTL, Q1B

R13 100k — Q1B, GND

SUPPLY 12 to 15 V DC, 2 A (18 V max)

J4 DC IN +/- — VCC, GND

C11 2200u 25V — VCC, GND

C12 0.1u — VCC, GND

PWR_FLAG — VCC

PWR_FLAG — GND

J7 HEATSINK GND (M3 lug) — GND

TRANSISTOR PINOUT FIELDS: columns L E B C, rows = legs 1..3

JQ1 Q1 2N3904 pinout field

GND	2			
Q1B	3			
Q1C	4			
	5			
GND	6			
Q1B	7			
Q1C	8			
	9			
GND	10			
Q1B	11			
Q1C	12			

JQ2 Q2 2N5401 pinout field

VCC	2			
PB	3			
MV	4			
	5			
VCC	6			
PB	7			
MV	8			
	9			
VCC	10			
PB	11			
MV	12			

Verified in ngspice (../sim/amp_full.cir): 30.3 dB, 20 Hz – 190 kHz, 4.4 W into 4R at 13.2 V before clipping, mute pin 6.9 mA.

RV1 is a dual B100K LINEAR pot (or 2 x X9C104 on J2 later: they need 5 V and CS/INC/U-D from the ESP32). R3/R4 give it a near-log law.

U1 heatsink tab to GND. Off-board: RV1/RV2 (dual pot), SW1, LS1, LS2. Not built yet – see docs/notes.html for the simulation and the unverified items.

Sheet: /
File: la4440.kicad_sch

Title: LA4440 stereo amplifier – volume header, toggle / GPIO mute

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.8		Id: 1/1